



Assessing the relative contribution of the facets of agility to distributed systems development success: an Analytic Hierarchy Process approach

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Abstract

Recent studies have sought to identify different types/facets of agility that can potentially contribute to distributed Information Systems Development (ISD) project success. However, prior research has not attempted to assess the relative importance of the various types of agility with respect to different ISD success measures. We believe that such an assessment is critical, since this information can enable organizations to direct scarce organizational resources to the types of agility that are most relevant. To this end, we use the Analytic Hierarchy Process to unearth, from the perspectives of two stakeholder groups of distributed software development projects, managers, and technical staff members, as to which agility facets facilitate (and to what degree) on-time completion of projects and effective collaboration in distributed ISD teams. Furthermore, noting that there is a need for an overall set of prioritized agility facets (by integrating managerial and technically oriented perspectives), we present three ways to aggregate the preferences of the two groups.

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Introduction

In the last few years, the use of distributed teams to accomplish different types of organizational work has increased exponentially (e.g., Saunders, 2000; Davison & de Vreede, 2001; Crowston *et al.*, 2005; Kotlarsky & Oshri, 2005), and is fast 'becoming [a] common practice in today's industry' (Layman *et al.*, 2006, p. 781). Distributed teams are typically assembled for the duration of a project by drawing on professionals situated in multiple locations (e.g., Jarvenpaa & Leidner, 1999; Sarker & Sahay, 2004). Distributed teams have been used in diverse areas such as disaster recovery (Altschuller & Benbunan-Fich, 2008), public sector inspectorates (Breu & Hemingway, 2004), and in the context of inter-organizational alliances (Maznevski & Chudoba, 2000), though their use is most prolific in the software development arena (Carmel, 1999; Mockus & Herbsleb, 2002). Distributed software development (DSD) projects have been specifically defined as 'projects that consist of two or more [sub] teams working

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together to accomplish project goals from different geographical locations' (Kotlarsky & Oshri, 2005, p. 37). Often, there are situations where at least one of the locations is in a different country, leading to globally DSD projects. Given the significant challenges faced by DSD teams, such as those related to differences in time, space, knowledge-level of team members and sometimes even culture and language, and a higher degree of reliance of the teams on asynchronous communication (e.g., Herbsleb & Moitra, 2001; Sarker & Sahay, 2004; Kotlarsky & Oshri, 2005; Layman et al., 2006), research on the effectiveness of such projects have received considerable interest (e.g., Ebert & De Neve, 2001; Sarker & Sahay, 2003; Kotlarsky & Oshri, 2005). Further, when we consider that 'over a quarter of large [ISD] projects are never completed', a majority of such projects are 'not completed on time or on budget', and even the ones that are completed, 'some three quarters of them are considered operating failures that do not function as intended' (Faraj & Sambamurthy, 2006, p. 238), it is clear that understanding the effectiveness of distributed ISD projects is critical.

Prior research on DSD projects have typically focused on technology or methodology-related factors that impact its effectiveness (e.g., Kotlarsky & Oshri, 2005). Specifically, Kotlarsky & Oshri (2005, p. 38) argue that the IS literature on DSD has focused mostly on 'technical aspects related to system development projects', with researchers restricting their attention to only understanding how the 'proper application of technical and operational mechanisms such as collaborative technologies, IS development (ISD) tools, and coordination mechanisms' affect DSD effectiveness. In recent years, however, the ISD process is being increasingly acknowledged as a complex behavioral/social process (Curtis et al., 1988; Newman & Robey, 1992; Lee & Xia, 2005), and the DSD process has been described as consisting of a set of interdependent ISD tasks that are undertaken by multiple individuals engaged in communicative transactions with one another within a team context that (at least initially) lacks in social integrativeness and mutuality (Sarker & Sahay, 2003). Thus, there is a greater need to explore the impact of social and behavioral factors on DSD effectiveness (e.g., Herbsleb & Mockus, 2003; Kotlarsky & Oshri, 2005).

Over the last few years, yet another trend has affected DSD research and practice. Realizing that DSD project teams 'need to react to market dynamics', respond and incorporate changing customer requirements, and also deal with everyday 'technological innovations', the concept of *agility* is increasingly being seen as relevant (Borjesson & Mathiassen, 2005, p. 360). In fact, many DSD projects and software organizations are adopting 'agility principles and approaches' in an effort to render them more effective (Borjesson & Mathiassen, 2005, p. 360). Many organizations involved in DSD (e.g., IBM Corporation, 2009) are openly acknowledging the importance of agility to their mission and heralding their

adoption of agility in their business processes. Consistent with the evidence in the industry, the broader literature on agility argues that agility leads to higher effectiveness for an organization (e.g., Sambamurthy et al., 2003; van Oosterhout et al., 2006; Lui & Piccoli, 2007; Martensson, 2007). For example, van Oosterhout et al. (2006, p. 132) state that agility enables organizations to 'adjust and act swiftly'. In the context of ISD, Fitzgerald et al. (2006, p. 200) argue that agility and agile methods can 'help address the key problems in software development'. However, in spite of this general interest in agility, and its acknowledged contribution to ISD success, there is limited empirical evidence of its effect on ISD (or DSD) effectiveness (Borjesson et al., 2006). In fact, Borjesson & Mathiassen (2005, p. 360) explicitly state that there is 'no research available that explores how these [agility] principles and approaches' affect DSD effectiveness. Our study takes a first step in addressing this void in the existing literature.

Agility has been defined in the literature in a variety of different ways, ranging from the *sense and response-to-change* capabilities (e.g., Overby et al., 2006; van Oosterhout et al., 2006) to the *use of different lightweight methodologies and practices* (e.g., Conboy & Fitzgerald, 2007). While there is little consensus on the formal definition of agility, it has been widely acknowledged that agility is not a monolithic construct, and is itself composed of *different facets* (e.g., Lee et al., 2006; Lui & Piccoli, 2007; Qumer & Henderson-Sellers, 2008a, b). For example, in the context of software development *methodologies*, Qumer & Henderson-Sellers (2008a) view agility being composed of a number of dimensions such as flexibility, speediness, etc. Further, Qumer & Henderson-Sellers (2008b, p. 1899) also propose the *Agile Software Solution Framework*, which highlights the importance of focusing on not only the agile methods, but also agile governance, and agile values in enabling organizations to adopt agile methods 'rapidly and effectively'. Recent studies have also sought to identify *different facets and dimensions of agility within the context of DSD teams* (e.g., Lee et al., 2006; Sarker & Sarker, 2008). However, we are not aware of an attempt to empirically assess the relative importance of the various facets of agility on the success of DSD projects. We seek to address this void in the current study.

We note that the literature has universally acknowledged that the two important stakeholders of ISD projects are the *technology workers* and the *project managers*, and that these two types of stakeholders differ significantly in terms of their orientation toward ISD (and hence DSD) projects (e.g., McMurtrey et al., 2002; Procaccino et al., 2005). For example, researchers contend that unlike the technology workers (e.g., developers), managers of ISD projects tend to overlook the operational and micro-interactional issues among the developers, and instead focus primarily on resource considerations (Procaccino et al., 2005, p. 190). Given these acknowledged differences among the managerial and technical staff of

projects, we seek to examine the following research question:

RQ1: *What are the different facets of agility significantly contributing to the selected success criteria of DSD teams, from the managerial perspective and from the technical perspective?*

While acknowledging the fact that technical and managerial personnel have different perspectives regarding how DSD teams should be organized, and that these groups are likely to prioritize agility facets differently, we feel that in many situations it may be meaningful to identify the key agility facets (and, if applicable, their relative priorities) for DSD teams as a whole, *taking the managerial and technical perspectives into consideration together*. Such an assessment is important, since both managerial and technical viewpoints are valuable, and a list of critical agility dimensions derived by integrating inputs from managerial and technical personnel can help the organization allocate resources toward enhancing specific agility facets of a DSD team in a balanced manner. This leads to the second research question:

RQ2: *Overall (i.e., incorporating managerial and technical perspectives), what are the different facets of agility significantly contributing to the selected success criteria of DSD teams?*

We attempt to answer the above research questions through the application of the Analytic Hierarchy Process (AHP) technique on data gathered from managerial and technical DSD project personnel. The AHP technique helps decision-makers assess the *relative* importance of the different attributes/dimensions on a particular outcome (e.g., Bagchi & Rao, 1992; Saaty, 1994a,b), and therefore enables more focused attention to be directed on the dimensions that are deemed most critical.

The rest of the manuscript is as follows: first, we provide a review of the literature on agility. Next, we provide an overview of the DSD success measures. This is followed by a discussion of our methodology, including the respondent profile, data collection techniques, our analysis procedure, and results. We continue with an elaboration of our results, followed by the identification of future directions. Finally, we conclude by describing the major implications of our study.

Agility: a multifaceted concept

van Oosterhout *et al.* (2006, p. 133) argue that while ‘much has been said and written on’ agility, a ‘consensus on a definition of agility has not yet emerged’. Agility was first introduced within the field of manufacturing (Meredith & Francis, 2000), and was defined as the ‘ability of enterprises to cope with unexpected changes, to survive unprecedented threats from the business environment, and to take advantages of changes as

opportunities (Zhang & Sharifi, 2000, p. 496)’ (Holmqvist & Pessi, 2006, p. 147). While initiated in manufacturing, the concept of agility has been adopted in other disciplines such as supply chain management, general management, and information systems as well. Within supply chain, agility is viewed as market sensitivity, process integrativeness, and virtuality (Christopher, 2000). Other researchers (e.g., Lui & Piccoli, 2007, p. 123) define agility as the ability to ‘detect new techniques’, and adapt those techniques to the organization. It is argued that agility is more than just ‘flexibility’, and reflects ‘readiness for action or change’ (Henderson-Sellers & Serour, 2005). In fact, business agility is viewed as an extension of flexibility, and is defined as the ability to ‘swiftly change businesses and business processes beyond the normal level of flexibility to effectively manage unpredictable external and internal changes’ (van Oosterhout *et al.*, 2006). Qumer & Henderson-Sellers (2008a, p. 281) define agility as a ‘persistent behavior or ability of a sensitive entity that exhibits flexibility to accommodate expected or unexpected changes rapidly, follows the shortest time span ...’, Mathiassen & Pries-Heje (2006, p. 116), along similar lines, define agility as being similar to ‘nimbleness’, which enables an organization to respond ‘quickly and gracefully to both expected and unexpected events in their environment’.

Within the IS discipline, Sambamurthy *et al.* (2003, p. 245) define agility as the ‘ability to detect opportunities for innovation and seize those competitive market opportunities by assembling requisite assets, knowledge, and relationships with speed and surprise’. Specifically, Overby *et al.* (2006) argue that agility incorporates both the ‘sensing and response capabilities’ of the firm, where these two capabilities share a ‘symbiotic relationship’.

ISD agility has been defined as the ability of ‘information systems development and deployment methods to swiftly adapt to the changing business requirements’ (Lee *et al.*, 2006, p. 50). Specifically, it refers to the capability to ‘rapidly develop and deploy systems ... to tap into emerging business opportunities’ (Lee *et al.*, 2006, p. 50). Conboy & Fitzgerald (2007) suggest that ISD agility refers to the presence of different lightweight methodologies and practices, which enable organizations and ISD project teams to handle change. Similarly, Qumer & Henderson-Sellers (2008a, p. 281) define agility in the context of software development as a ‘method that is people-focused, communication-oriented, flexible ..., speedy, lean ... responsive, ... and learning’.

While there are clearly several definitions of agility, the above discussion suggests that there is consensus among researchers in terms of the *core elements of the conception of agility*, at least in the context of ISD. They are (1) a sensing capability, (2) a response capability to a wide range of unprecedented (and emergent) changes related to resources, environmental, or business process-related issues, and (3) the use of lightweight methodologies and practices. Our perspective to agility is consistent with this above view.

Our literature review also revealed that *agility* is itself composed of different components. Given its roots in

agile manufacturing, initially, agility was thought to be composed of two different dimensions: strategic agility, and operational agility. Meredith & Francis (2000) developed the 'agile wheel' composed (similar to a wheel) of four different quadrants (or dimensions) of agility. They are strategic agility incorporating properties such as 'wide-deep scanning, strategic commitment, full deployment, etc.', process agility referring to capabilities such as fast, new product development and acquisition, 'rapid decision-making', linkage agility with properties such as flexible partnerships, deep customer insight, and people agility referring to components such as 'multi-skilled, flexible people', 'continuous learning', etc. Along similar lines, Lui & Piccoli (2007, pp. 125–126) propose that information systems agility (owing to its being socio-technical in nature) is also composed of four different components. They are technological agility (the degree of flexibility of the information technology, and the extent to which it is capable of rapid adjustment whenever necessary), process agility (flexibility of the firm's business processes, and its ability to change rapidly with varying firm strategies), people agility (varied and multi-skills of individuals with the ability to be easily redeployed when faced with changes), and structure agility (such as 'workforce empowerment, distributed decision-making authority, and flatter managerial hierarchies'). While comprehensive in nature, the components discussed above apply to information systems in general, and are not specific to information systems development, the context of our study.

In the context of ISD, Lee *et al.* (2006) propose that agility in such teams is composed of agile IT strategy, agile IT infrastructure, and agile project management. While the components identified by Lee *et al.* (2006) provide a good start toward our understanding of agility in distributed ISD teams, it must be noted that their focus has been more on the macro issues surrounding distributed ISD (e.g., strategy, overall project management). On the other hand, Qumer & Henderson-Sellers (2008b) in their four-dimensional framework (4-DAT) view agility in the context of a software development method as being composed of 'flexibility, speed, leanness, learning, and responsiveness'. While immensely valuable, the components identified by Qumer & Henderson-Sellers (2008a, b) focus primarily on the *development methodology* used by software development teams. It must be noted that much of the success of distributed teams engaged in ISD relies on a wide range of both macro and micro issues such as its day-to-day operations, its application and use of ideal methodologies, and on the management of its resources and its relationships (e.g., Saunders, 2000), which need to be factored in as well.

A recent study (Sarker & Sarker, 2008), taking the above issues into consideration, and based on an intensive study of DSD within one of the premier technology companies in the world, proposed a more integrative set of the facets of DSD agility (see Figure 1). This study argues that agility in the context of DSD has three broad categories: (1) resource (agility arising from the DSD

team's access to necessary resources), (2) process (agility arising from the appropriate management of the process of DSD), and (3) linkage (agility arising from the nature of the relationships and communicative ties between the different DSD team members). Further, *resource agility* is composed of agility arising from accessibility to both people-related and technology-based resources. *People-related agility* refers to (a) the ease (and speed) with which the DSD team is able to acquire skilled human resources at each distributed location as and when necessary, and (b) the capability of the DSD team to rapidly (and gracefully) reconfigure itself on demand, by rotating and shifting different team members (across locations or divisions) in different roles and responsibilities, or by shifting the 'control centers' of the project from one distributed location to another. On the other hand, *technology resource-based agility* refers to both the extent to (and ease with) which the DSD team has access to high quality communication, development, and testing-related technologies, irrespective of their location.

Moving to the second broad category, *process agility* is seen to be composed of methodology-based agility, environmental scanning-based agility, and work-transition-based agility. *Methodology-based agility* refers to the ease with which the DSD team is able to gracefully adapt the use of traditional Systems Development Life Cycle (SDLC) and agile methodologies and practices to its project context, and also deal comfortably with methodology changes (if such changes are mandated) during the project. *Environmental scanning-based agility* reflects the systemic ability of the DSD team to anticipate, recognize, and react to changes in the project due to changes in the environment (e.g., changes in the external client organization or their requirements). *Work transition-based agility* refers to the seamless transition of work from one location to another around the world.

Finally, the third broad category of *linkage agility* is composed of temporal agility, cultural agility, and communicative agility. *Temporal agility* reflects the capability of the DSD team to work and collaborate with each other without significant temporal delays. Similarly, *cultural agility* reflects the ability of the DSD team to rise above any cultural differences that may exist within the team, and work seamlessly. *Communicative agility* refers to the ability of the DSD team to be aware of the actions/reactions of distributed team members, utilize cultural difference advantageously (where possible), and maintain continuous 'connectivity' with the customers.

Given the comprehensiveness of the above-mentioned facets of agility, we adopt these in our study, wherein we examine the importance of their roles in enhancing DSD project success from the perspective of industry practitioners. Next, we discuss DSD project success in further detail.

Information systems development project success

It has been widely acknowledged that a project is considered to be successful 'when it satisfies project

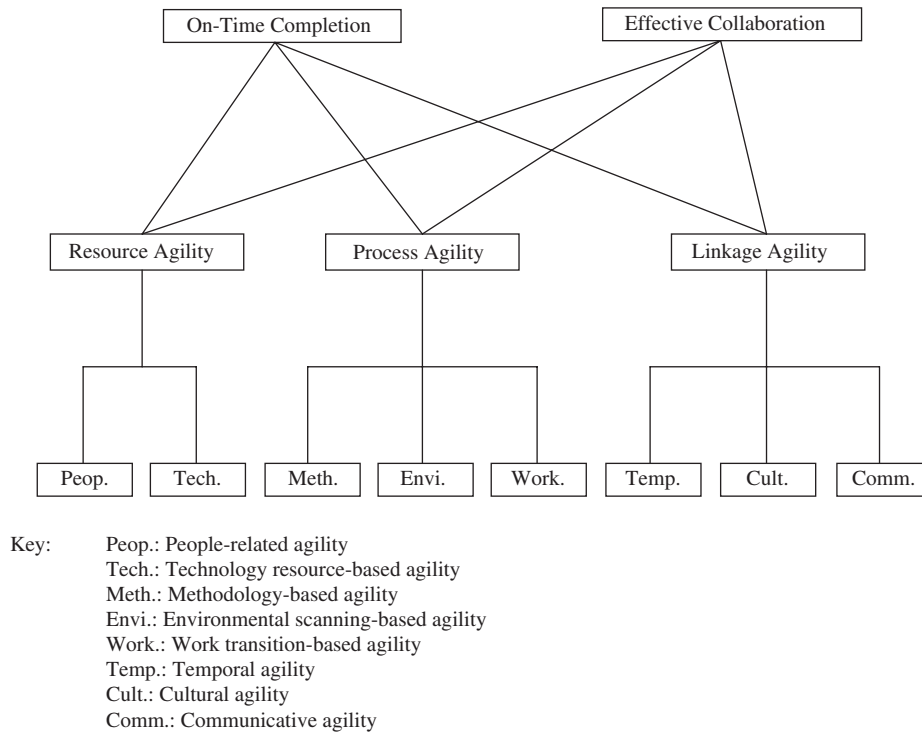


Figure 1 The hierarchy diagram.

objectives' (Nicholas, 1989, p. 24). Most projects have multiple objectives such as 'time, cost, quality, and safety' (Nicholas, 1989, p. 24). In the context of ISD projects, Jiang & Klein (1999) argue that cost (or budget) and on-time delivery or completion should be viewed as the core measures of ISD success. Phan *et al.* (1995) report, based on their study of software project management professionals, that such professionals typically view on-time delivery of an ISD project as a key measure of software project success. In addition to on-time delivery, other measures of ISD project success have been adopted and used, such as the quality of the system being developed, the improvements in the quality of the decision-making, IS usage, increased productivity in the organization (such as the lowering of cost, and increase in benefits) (e.g., Procaccino *et al.*, 2005). However, increasingly, it is being argued that the above-mentioned measures of ISD success are limiting, and do not capture the challenges that ISD project teams face in terms of 'communication, coordination, learning, and negotiation' (Faraj & Sambamurthy, 2006, p. 238). In fact, Procaccino *et al.* (2005, p. 188) argue that quantitative measures such as on-time delivery and schedule 'can be padded to create an illusion of success', and found from their survey of several ISD professionals, that many of them see 'value and success in a project that is delivered late'.

What the above discussion suggests is that ISD project success is a 'multidimensional concept' (Jiang & Klein, 1999, p. 264). On the one hand, there is the *product perspective* of ISD project success, which includes measures

such as meeting cost and schedule requirements of the project, and on the other hand, is the *process perspective*, which includes measures such as user satisfaction, and the project stakeholders' satisfaction with the development process (Procaccino *et al.*, 2005). The measures of the process and product perspectives are 'intertwined' and any study on ISD projects should include both (Procaccino *et al.*, 2005). Saarinen (1996), in an effort to provide such a comprehensive measure of ISD project success, proposed a scale that captured success with respect to (1) the development process, (2) the use process, (3) the quality of the information system being developed, and (4) the effect of the new information system on the organization. Faraj & Sambamurthy (2006, p. 238) also emphatically argue that examination of the effectiveness of modern-day ISD projects 'must include an understanding of the development process and "peopleware" '.

The importance of assessing an ISD project from the lens of process is perhaps nowhere more applicable than in the context of DSD, where the focus has traditionally been on the technical aspects (e.g., on-time), with little attention being paid to 'human and the social aspects' (Kotlarsky & Oshri, 2005). DSD projects suffer from several handicaps such as time and culture differences, distance between team members, reliance on computer mediated communication technologies for communication, coordination and knowledge sharing problems, and success of such projects should partly be assessed by examining how the project teams are able to rise above these challenges, and experience high levels of process

satisfaction or 'successful collaboration' (Kotlarsky & Oshri, 2005). In fact, in the context of DSD, it has been argued that researchers/practitioners should not only assess the product-based success measures (e.g., on-time delivery), but also assess process-based (or people-related) success measures (e.g., Gallivan, 2001) such as the collaboration effectiveness or the extent to which the project meets the 'psychological needs of the members' (Kotlarsky & Oshri, 2005).

Consequently, in this study, as a starting point, we view *DSD project success* as being composed of one product-based measure (i.e., *on-time delivery*), and one process-based measure (i.e., *effective collaboration*). We define 'on-time delivery' as the extent to which the ISD project has delivered the new (or enhanced) information system within the schedule promised to its clients. Drawing on Kotlarsky & Oshri (2005), we define 'effective collaboration' as the extent to which team members have experienced 'personal satisfaction', have been motivated to 'continue their engagement in collaborative work despite geographical, time, and cultural differences', and are motivated to 'engage in future teamwork' with the same members. Jiang & Klein (1999), in the context of their study on the effect of different types of risk on ISD project success, argued that in order to provide actionable guidance, it is important to examine the separate role of each of the different types of risk separately on the different dimensions of project success. In this study, we follow their guidelines by examining the different facets of agility (discussed earlier) *separately* on the 'on-time delivery' and 'effective collaboration' measures of DSD project success.

Research methods

Our discussion of agility above indicates that agility in the context of DSD teams is composed of multiple facets, with each facet further divided into sub-facets (e.g., resource agility is comprised of people and technology-based agility). Given the number of facets of agility and the infeasibility of focusing equally (and simultaneously) on all of them, unless there is an assessment, or at least a framework for assessing the relative importance of the different agility elements (Bagchi & Rao, 1992), it would be extremely difficult for decision-makers to decide on the specific agility dimensions they should attempt to enhance in a DSD project team.

Saaty (1994a) argues that, in such contexts, the AHP can be a helpful technique. The AHP technique enables decision-makers to assess the relative importance of the different attributes on a particular outcome (Bagchi & Rao, 1992) and thus, direct more focused attention to the alternatives that matter the most to an outcome. In our study too, while our eight proposed facets of agility have emerged from past research, to provide more guidance to decision-makers in charge of DSD teams, it is important to determine which of the facets need to be considered to achieve project success. Knowledge of relative importance can assist decision-makers by providing guidance

as to where to invest time and resources for agility improvement, and thereby maximize the probability of fostering successful projects in the DSD environment. Thus, we believe that the use of AHP as the analysis technique is an interesting and productive way forward.

The AHP technique was developed by Thomas L. Saaty (Saaty, 1980) as a method that helps decision-making, especially when there are multiple factors, and the decision-maker does not have solid *a priori* importance weights to place on each of the factors. A hierarchy of the factors associated with the decision-making context is generated, and via pair-wise comparisons of the factors in each level of the hierarchy, factor weights are generated. Various alternatives are then evaluated using these weighted factors, and the alternative yielding the highest total weighted score is deemed the best. Normally, the pair-wise comparisons in AHP are provided as a single set by one expert/manager or a committee of decision-makers (Saaty, 1994b). For our purposes, however, we were concerned that results based on comparisons provided by just one IS expert may reflect this expert's experiences – this would be problematic if the expert's experiences had been unique due to some reason. Therefore, even though obtaining pair-wise comparisons of various factors is mentally taxing on the practitioners, and also expensive with respect to time required, we went beyond the norm and obtained comparisons from eight DSD experts, four technically oriented, and the other four managerially oriented.

Data collection

Our first step was to identify the experts who were capable of providing (and willing to provide) the pair-wise comparisons. In selecting our eight experts, we paid attention to the roles they typically play within a distributed ISD project, using a mix of technical-focused and managerial-focused DSD experts in the spirit of theoretical sampling. Once the identified experts agreed to participate in the study, they were provided with the definitions of the different facets of agility for familiarization. Through in-depth interviews conducted by three of the authors (which typically lasted between 1 and 1½h), the experts provided the relative importance of each element of agility over each other element of agility on the success measures of distributed ISD projects adopted in this study. Table 1 provides details about our respondents. As shown in this table, our experts represented a number of different industries, ranging from technology and consulting firms to the U.S. federal government. Four of our experts were in managerial roles, while our other four primarily played technical roles within their DSD projects.

The AHP procedure

Saaty (1994a) suggests that within the AHP technique, a problem needs to be structured as a 'hierarchy', where the top level consists of the 'objective of the problem' (Bagchi & Rao, 1992, p. 92). In the next level, the sub-objectives

Table 1 Respondent profiles

Respondent number	Company affiliation and its HQ	Technical or managerial role?	Country currently based in	Years of experience in distributed IS contexts	Number of distributed projects	Agile ISD methods (e.g., XP, Scrum) experience	National contexts of distributed projects
R1	Infosys (India)	Technical	U.K., Switzerland	10	10	Yes	India with Japan, Singapore, Australia, South Africa, and Europe (including the Netherlands, the U.K., and Switzerland)
R2	Microsoft (U.S.)	Technical	India	8	12	Yes	U.S.–India
R3	Intec Telecoms (U.K.)	Managerial	India	12	100	Yes	U.S.–U.K.–India, U.K.–Australia–India–Poland, U.S.–U.K.–Japan–India
R4	Deloitte Consulting (U.S.)	Technical	U.S.	2	2	Yes	U.S.–India
R5	Federal Government (U.S.)	Managerial	U.S.	10	4	Yes	Different parts of the U.S.
R6	Agilent (U.S.)	Managerial	U.S.	4	4–5	No	U.S.–India
R7	Cap Gemini (France)	Managerial	U.S.	7.5	12	Yes	U.S.–China, U.S.–India
R8	EMC Corp. (U.S.)	Technical	U.S.	10	15	No	Different parts of the U.S., U.S.–Russia, U.S.–India, U.S.–Israel

or elements playing a role in achieving the objective(s) are placed. This level is followed by the placement of any criterion variables (or lower-level objectives) creating the higher-level objectives. As discussed earlier, we chose *on-time delivery and effective collaboration* as the two primary project success factors (i.e., objectives) for our purposes. The next level of the hierarchy has our three broad categories of agility influencing each of the project success factors. Finally, the bottom level of the hierarchy incorporates each of the facets (i.e., sub-categories) of agility feeding into its respective agility categories. Figure 1 illustrates our AHP hierarchy.

Once the hierarchy is established, the ‘prioritization’ process ensues, where the experts provide judgments (through a pair-wise comparison) about the relative importance of all possible pairs of elements on the final objective. Subjects provided their numerical pair-wise comparison ratings using the traditional AHP rating scale (Saaty & Vargas, 1982) (please see Table 2).

A relative importance matrix is created for each set of pair-wise comparisons (i.e., for each respondent). We provide a sample relative importance matrix in Table 3.

To accommodate the differing opinions of each expert pool (i.e., technical and managerial), a composite matrix is generated for each pair-wise comparison set. It is computed by taking the geometric mean of the respective cells in the respondents’ matrices within each group (Saaty & Vargas, 1982). Such a strategy allows the judgments and their reciprocals to be viewed symmetrically, as the reciprocal of the geometric mean of a set of judgments is the same as the geometric mean of the reciprocals.

The next step involves the creation of the priority vector of rankings of the different agility facets. Specifically, scores from the relative importance matrices are normalized to create relative weights within the comparison set, as well as overall weights for the entire hierarchy.

Consistency

A natural problem that arises when asking humans to compare factors one by one by assigning an importance score is the issue of logical consistency in their responses. For example, if factor A is slightly favored over factor B and B is slightly favored over factor C, then we would expect A to be strongly favored over C. In this case, if the respondent were to say that A was only slightly favored over C, A was extremely favored over C, or, worse yet, C was favored over A, the responses are said to contain a certain amount of inconsistency. It is important to note that consistency is an ideal; however, because of the salience of one factor over another in the respondents’ minds, sometimes, due to recent experience/exposure, a set of pair-wise comparisons often does show inconsistency. AHP incorporates a *consistency ratio* to help detect this phenomenon, in which higher ratio values imply higher inconsistency. When working with responses from only one decision-maker, comparisons that are too inconsistent usually cause the AHP researcher to

Table 2 Rating scale

1 (equally important)	3 (moderately important)	5 (more strongly important)	7 (very strong or demonstrated importance)	9 (extremely important)
Two activities contribute equally to the objective	Experience and judgment slightly favor one activity over another	Experience and judgment strongly favor one activity over another	An activity is favored very strongly over another, its dominance demonstrated in practice	The evidence favoring one activity over another is of the highest possible order of affirmation

Alternately, you may also assign ratings of 2, 4, 6, or 8 as a compromise between the above values.

Table 3 A sample relative importance matrix

Type of agility	On-time delivery		
	Temporal linkage-based	Cultural linkage-based	Communication linkage-based
Temporal linkage-based	1	1/3	1/7
Cultural linkage-based	3	1	1/5
Communication linkage-based	7	5	1

request the respondent to re-score his or her comparisons. Typically, inconsistencies around 10% are considered reasonable (e.g., Saaty, 1980; Render *et al.*, 2006). In our case, we discarded the eight out of 72 responses that had a high inconsistency ($> 12\%$), and we computed the geometric mean of the remaining consistent scores. The maximum consistency ratio of the resulting composite matrices (covering managerial role, technical role, and both roles) was 5.5%, well within the 10% cutoff value.

Results

As noted earlier, in light of the socio-technical nature of DSD (Sarker & Sahay, 2004), we invited an equal number of 'technical' (Subjects 1, 2, 4, and 8) and 'managerial' (Subjects 3, 5, 6, and 7) respondents (see Table 1). Virtually every DSD team has both *technical* and *managerial* members (e.g., McMurtrey *et al.*, 2002; Procaccino *et al.*, 2005), and while both groups work collaboratively toward the success of the project team, their perspectives with respect to various aspects of a DSD project, including the relative importance of agility facets, are often different. Further, the agility facets that are associated with on-time completion (and the priorities of these facets) can be expected to be different from those associated with effective collaboration. Recognizing the potential differences between managerial and technical perspectives on effective collaboration and on-time completion, we conduct our analysis separately for these four combinations. Table 4 displays the weights produced from the AHP analysis of the different facets of agility with respect to the two different success measures of DSD projects for the two groups of respondents, managerial and technical. Each weight reflects

the importance of that element to the specific success measure. The weights should be interpreted as relative to each other.

At this point, it may be useful to explain the basis of the cutoff value used to identify important facets of agility within each cell. If all eight sub-dimensions were deemed equally important, then all weights would equal 0.125 (i.e., $1/8$). Thus, the lists of categories with weights attached can be interpreted in two ways: (1) categories are ranked from most important to least important, and (2) *scores above 0.125 demonstrate a higher-than-average weight* (our criterion for significance here) while scores below 0.125 demonstrate categories of lower significance. For example, in the context of this study (see Table 4), a glance at the cell pertaining to the managerial perspective on on-time completion suggests that the *people-based* and *technology-based agility* are of significance. Similarly, *work transition*, *temporal*, *methodological*, and *people-based agility* are of significance for achieving on-time completion, from the point of view of technical respondents. Along similar lines, effective collaboration is associated with high level of *communicative* and *people-based agility* by managerial respondents, whereas technical respondents view *communicative*, *cultural*, *environmental*, *temporal*, and *work transition-based agility* as especially important to achieve effective collaboration.

In Table 5, we present key agility facets for on-time completion and effective collaboration aggregated across technical and managerial perspectives. While the technical and managerial perspectives are different, from the organizational standpoint, it may sometimes be important to know what facets of agility *overall* in a DSD team deserve attention for ensuring that on-time completion of the project (or effective collaboration) is achieved.

Table 4 Relative importance of agility facets for on-time completion and effectiveness of collaboration across managerial and technical perspectives

<i>ISD success criterion</i>	<i>Managerial perspective</i>	<i>Technical perspective</i>
On-time completion	People (0.286) Technology (0.272) <i>Weights below the cutoff of 0.125</i> Communicative (0.121) Temporal (0.116) Methodological (0.066) Work Transition (0.058) Environmental (0.040) Cultural (0.040)	Work Transition (0.258) Temporal (0.206) Methodological (0.156) People (0.137) <i>Weights below the cutoff of 0.125</i> Technology (0.095) Communicative (0.071) Environmental (0.048) Cultural (0.030)
Effective collaboration	Communicative (0.280) People (0.171) <i>Weights below the cutoff of 0.125</i> Cultural (0.105) Methodological (0.103) Work Transition (0.103) Temporal (0.099) Technology (0.093) Environmental (0.047)	Communicative (0.226) Cultural (0.153) Environmental (0.142) Temporal (0.137) Work Transition (0.126) <i>Weights below the cutoff of 0.125</i> Methodological (0.079) People (0.074) Technology (0.063)

Note: Agility facets with weights greater than 0.125 are considered to be of significance and are thus represented using bold text.

Table 5 Key agility facets for on-time completion and effective collaboration aggregated across technical and managerial perspectives

<i>Aggregation approach for integrating managerial and technical perspectives</i>	<i>Facets resulting from aggregation using 'geometric mean' (with weights)</i>	<i>Facets resulting from aggregation using set-theoretic operator 'union'</i>	<i>Facets resulting from aggregation using set-theoretic operator 'intersection'</i>
<i>Success criteria</i>			
On-time completion <i>Agility facets</i>	People (0.203) Technology (0.162) Temporal (0.158) Work Transition (0.135) Methodological (0.127) Communicative (0.114) Environmental (0.061) Cultural (0.042)	People Technology Work Transition Temporal Methodological	People
Effective collaboration <i>Agility facets</i>	Communicative (0.259) Cultural (0.131) Work Transition (0.121) Temporal (0.117) People (0.114) Methodological (0.093) Environmental (0.086) Technology (0.079)	Communicative People Cultural Environmental Temporal Work Transition	Communicative

Note: The agility facets that emerged as significant through the aggregation process are represented in bold.

One approach, of course, is to numerically aggregate (using geometric mean as before) across the two groups of respondents (technical and managerial). While this is a useful approach, a limitation is that if there is very high degree of heterogeneity in the preferences of the two groups, such an aggregation may lead to misleading results (e.g., Klein *et al.*, 1994). Another approach is to logically aggregate the managerial and technical perspectives using the set-theoretic operator of union ($\cup$) (Childress, 1974). For on-time completion, we compute {People, Technology} $\cup$ {Work Transition, Temporal, Methodological, and People} (see Table 4, row pertaining to on-time completion), which gives us {People, Technology, Work Transition, Methodological}. In other words, using the union operator, we are able to identify the set of agility facets that the organization needs to pay attention to, assuming that the managerial as well as the technical perspectives are considered to be valuable. Yet another approach of logical aggregation is by using the set-theoretic operator of intersection ($\cap$) (Childress, 1974). Again, for on-time completion, we compute {People, Technology} $\cap$ {Work Transition, Temporal, Methodological, and People} (see Table 4, row pertaining to on-time completion), which gives us {People}. In other words, using the intersection operator, we are able to identify the agility facet(s) that is (are) considered significant from *both* the technical and managerial perspectives. Using a similar approach, and considering the row pertaining to effective collaboration in Table 4, we aggregate the technical and managerial perspectives on achieving effective collaboration. Interestingly, the top-weighted geometric mean of all respondents under each success criterion matched its corresponding 'intersection' facet. An 'intersection' approach to aggregation may be valuable if the organization has very few resources available for enhancing DSD agility, and the decision-maker wishes to focus on the facet of agility seen as significant from the technical as well as the managerial perspectives. While we have presented three possible ways of aggregating results derived from the inputs of managerial and technical respondents, and outlined their advantages, certainly other approaches exist, and we invite future work in that direction.

Discussion of results

In this section, we revisit the results, make sense of the numbers, and unearth key implications. As we have noted earlier, the primary goal of the study is to gain understanding on the relative importance of the different facets of agility, from the point of view of practitioners actively engaged in ISD within a distributed setting. As mentioned earlier, we first sought to compare the managerial and technical perspectives with respect to the perceived contribution of the different agility elements on on-time completion and effective collaboration. Table 4 shows the relative importance of the facets of agility, from the point of view of respondents with

primarily a managerial vs a technical role in the DSD projects.

In order to appreciate the similarities and differences among the four cells, it is useful to develop an understanding of the managerial vs the technical perspectives in DSD. Based on our interactions with our respondents in a managerial role, we found that managers are fundamentally concerned with ensuring appropriate resources exist for the project, and managing activities and configurations to harness the resources (e.g., Procaccino *et al.*, 2005). Technical workers, on the other hand, being intimately involved in efforts everyday for efficient and effective task completion, appeared to be much more concerned with operational aspects, considering resource issues as necessary but not of paramount importance.

Looking at Table 4, specifically the row corresponding to on-time completion, we see that indeed managers consider *people-based* and *technology-based agility* as significant factors. The results indicate that managerial perspectives value access to the most important resources for ISD, people-based resources, that is, individuals with requisite skills, whom they can bring into the project at any given location within a short period of time. This could include the ability to relocate relevant individuals from one site to another, and having access to flexible resource pools at different sites (i.e., talent available for hiring, talent available from other parts/project-teams within the organization, or access to vendors with strong 'bench strength'). The importance of people (at least from the managers' points of view) is not surprising given that prior research and practice surrounding software development has consistently argued that people are the most important components of such projects (McConnell, 1996). Managers also see their role as providing their team access to a set of suitable technological resources – communication and collaboration infrastructure that can be accessed in a locationally transparent manner, appropriate development tools and shared server resources for efficient development, testing, version control, etc., and multiple backup sites with recovery capabilities to protect against political and natural catastrophic events. This is consistent with existing research as well, where it has been found that managers are often obsessed with the technical issues of the project, especially 'those related to hardware and software (such as programming language and compilers)' (e.g., Procaccino *et al.*, 2005, p. 190).

While managers tend to have somewhat of an outside view into the distributed ISD process, consistent with the literature, the technical team members, at least in our study, tended to have more of an inside view to the team's processes, with a focus on the operational components of the DSD project. Consequently, *work transitions* (i.e., how work was decomposed/distributed across locations, the extent to which there was ease of handing off work across locations around the world, and the feasibility of productively accomplishing tasks around the clock in the so-called 'follow the sun' mode), and *temporal agility* (i.e., the ability to minimize unproductive

waits resulting from temporal distances) emerged as the two most important. Ramesh *et al.* (2006, p. 43) comment that 'While 24×7 development [i.e., follow-the-sun approach] is sometimes claimed to be a benefit of distributed development, this was far from reality ...'. In fact, proper transition of work is seen as one of the biggest barriers to the successful completion of distributed projects. It is thus not surprising that for the technical members of the team (the ones who are involved in the day-to-day task accomplishment), effective work-transition was viewed as the most formidable contributor to on-time completion. Similarly, time delays are especially critical in software development that involves interdependent tasks performed by 'people at more than one site' (Herbsleb & Mockus, 2003, p. 482). Delay due to time differences becomes a major impediment to on-time delivery in at least two ways: (1) in the context of teams with members distributed in locations with minimal working-time overlaps, in the absence of tactics to bridge the time differences, team members need to wait several hours before clarifications can be received, or problems are resolved, and this may hold up work (e.g., Herbsleb & Mockus 2003; Sarker & Sahay 2004); (2) often, by the time clarifications are received (sometimes this takes several days before the precise answer is received), the work may have progressed based on assumptions made by the side requesting clarifications. If the assumptions made do not match with the actual responses provided by the other side, much work needs to be redone, which in turn leads to additional loss of time. It is only natural that especially developers feel frustrated when having to redo 1 or 2 days of work, and thus, they start avoiding asking questions. Of course, this does not solve the problem, but actually aggravates it, because the lack of correspondence between the work accomplished and the expectations come to light at a later stage, and this can lead to even greater delays in completing the project (e.g., Sarker & Sarker, 2008).

Setting up and adhering to appropriate methodologies was also found to be important, given that methodologies provide the structure for effective and flexible utilization of people and technology. In fact, it has been argued that 'poor software development [methodological] practices' place significant 'risk' on a software development project (Procaccino *et al.*, 2005, p. 190). Finally, the importance of having qualified personnel available for quick deployment in any given location also featured as an important criterion from the technical perspective (as well as from the managerial perspective).

With respect to effective collaboration, communicative agility was seen as important by both managers and technical respondents. This is understandable since communication is at the core of any human connection that is essential to collaboration, and is consistent with prior research. Herbsleb & Mockus (2003, p. 481), in fact, argue that 'in sharp contrast to the popular image of software developers [i.e., the technical people] as relatively introverted and isolated, they, in fact, spend a large

proportion of their time communicating'. Thus, having multiplicity of linkages among distributed members, and between the DSD team and their client(s) was seen as valuable. As discussed earlier, managers often have a more outside-in view to the team dynamics, and focus more on resource issues (e.g., Procaccino *et al.*, 2005). Thus, they tend to see access to appropriate human resources as contributing to effective collaboration. A possible explanation is that if sufficient number of skilled personnel is not available for rapid deployment when necessary, the team members tend to get stretched through overwork, affecting their work-life balance and consequently, the collaborative relationships among them tend to suffer. On similar lines, if personnel are relocated from one location to another, unless they can gracefully adapt to the new social systems with unique work practices, the nature of collaboration can be negatively affected.

Apart from communication, the two other facets of agility found important by those with a technical perspective were cultural agility and environmental agility, both of which find no place (or lower places) in the other lists. Cultural agility, arising from team members' cultural awareness and sensitivity of other team members and clients, in terms of both national culture and workplace culture, is an obvious contributor to effective day-to-day collaboration across geographical locations, especially across geographical boundaries (e.g., Carmel, 1999). Lack of cultural sensitivity can lead to breakdowns in collaborative relationships; for example, different assumptions regarding work and IT, dates of (and social activities associated with) religious holidays, etc. can all contribute to the lack of seamless human connection (e.g., Herbsleb & Mockus, 2003; Sarker & Sahay, 2004). The inclusion of environmental agility as an important factor for effective collaboration from a technical perspective is a bit perplexing. Perhaps what is being indicated is that technical personnel have to bear the brunt of any changes that occur in the environment (e.g., change in strategy within the client organization, turnover in vendor organizations) and, thus, they place high value on environmental scanning and planning accordingly, so that that steady seamless collaboration can be sustained. Finally, temporal agility and work transition continued to be viewed as important for an effective collaboration from the technical perspective.

Table 5 shows the relative importance of the various facets of agility with respect to on-time completion and effectiveness of collaboration, across our entire set of eight experts, using three different aggregation techniques. Using the aggregation of the 'geometric means', and the set-theoretical operator 'union' techniques, we obtain fairly similar results with respect to on-time completion. Specifically, people-based agility, technology-based agility, temporal agility, work transition agility, and methodological agility contribute to on-time completion, while communicative agility and cultural agility are key contributors to effective collaboration. The first

observation that may be made is that on-time completion requires five different facets of agility, encompassing human, technological, process, and linkage considerations. Interestingly, the five factors, especially the last four, have comparable levels of importance (based on weights shown), which possibly indicates that for on-time completion, the teams need to agile on many fronts, and lack of attention to any front may result in slippage as far as meeting deadlines is concerned. This is consistent with recent views surrounding the critical success factors affecting on-time completion of ISD projects. For example, Chow & Cao (2008, p. 963) note that a large number of factors ranging from highly skilled people with 'competence and expertise', proper technological infrastructure, good communication among the stakeholders, following a systematic methodology for software development, proper coordination of work are all required to complete a project on time. Interestingly, the 'intersection' results highlight, that the single most important agility facet contributing to the on-time completion of projects as viewed by both the managerial and technical members is *people-based agility*. This result further reaffirms the view that in the context of DSD projects, people are the most important components of such projects (e.g., McConnell, 1996; Lee *et al.*, 2006).

In contrast, for effective collaboration, communicative and cultural agility (as per the aggregation using 'geometric means'), and communicative, people, cultural, environmental, temporal, and work transition (as per the 'union' technique) emerged as the most important. Finally, the single most important contributor to effective collaboration, as viewed by both the managerial and technical members (which emerged through the 'intersection' technique) was 'communicative' agility. This result is not surprising since, for seamless collaboration, the quality of communicative linkage among the team members, and with important stakeholders – notably the clients – is key (e.g., Herbsleb & Mockus, 2003). What this result suggests is that if a DSD project team experiences resource crunch, the aspect of agility that they need to focus on in an effort to ensure an effective collaboration should be communicative agility.

However, if the team is in a position to manage (and juggle) multiple dimensions of agility at the same time, then, along with communicative agility, people-based agility, cultural, environmental, temporal, and work-transition based agility should also receive attention. With respect to people-based agility, one could argue that if a sufficient number of skilled personnel are not available for 'ramp-up' when necessary, the team members tend to get stretched through overwork, and the relationships among them tend to suffer; moreover, if personnel are transferred from one location to another, unless they can gracefully adapt to the new environments, in terms of decision-making and working styles, the nature of collaboration can be negatively affected. In the end, collaboration is all about people working together and about the human connection with high

situated awareness among parties, whether co-located or distributed (e.g., Sarker & Sahay, 2003), and so it is reasonable to find the relevant agility facets to be more people driven overall. Similarly, given the cultural diversity of DSD teams, management of the heterogeneous cultures, and the ability of members to transcend beyond their cultural differences and work cohesively as 'one' team is 'critical' for the successful collaboration (Lee *et al.*, 2006, p. 51). Furthermore, unless the DSD team possesses the ability to scan the environment and respond to changes rapidly, to manage the complications associated with different time zones, and finally, to smoothly transition work from one location to the other, it is clear that effective collaboration is difficult to attain.

Future directions

Our research lays a foundation upon which further research can be conducted. For example, given the subjective nature of the responses and the dependence upon a sample size of eight experts, confirmatory research using other and/or more respondents could help to support our conclusions (e.g., Baskerville & Lee, 1999).

While we feel fairly confident about the methodological rigor of our study and the results that we have obtained, we are also of the opinion that exploration of any phenomenon in any great depth can only be achieved through methodological pluralism (e.g., Lee, 1991). We therefore feel that there should be further investigations about the connection between agility and DSD success and the relative importance of agility facets using other methodological approaches that would complement and/or supplement our approach. One interesting approach could be the use of an in-depth interpretive case study that focuses on particular DSD projects, and examines the presence (or absence) of the particular facets of agility and their relationship with project success. Another (more positivist) approach would be to propose a theoretical research model with formal hypotheses based on the findings of the current study and then test it through empirical investigation (possibly a survey) involving quantitative data and statistical analytic techniques. As our study is primarily based on the subjective understanding of practitioners who are immersed within the phenomenon in question, the follow-up interpretive and positivist studies suggested above would provide an opportunity to implement an integrative agenda based on the framework proposed by Lee (1991). His framework proposes the investigation of a phenomenon through the perspectives of three levels of understanding – subjective, interpretive and positivist, and a cyclical relationship between them. In this framework the subjective understanding provides the basis for the interpretive understanding, which in turn provides the basis of the positivist understanding that can lead to the formulation of predictions about the phenomenon in question.

Finally, in this study we have utilized just two of the many dimensions of success as our objectives. While we believe that the two are very relevant in the DSD context, clearly other success measures such as completion within budget, quality of product, and client satisfaction could have been studied as well, and we invite future research on these lines. Similarly, we have attempted to compare and contrast priorities of respondents with technical vs managerial roles. Other interesting comparisons could be based on the nature of work arrangement contexts (insourcing vs outsourcing; onshoring vs nearshoring vs offshoring), and based on location (developed vs developing country), etc. While we have only taken a first step, we still believe that we provide a potentially relevant set of findings, and a useful approach to investigate issues of this nature in the arena of DSD.

Conclusion

While there is a growing recognition that the adoption of agility principles can enhance the success of software development in general, and in the DSD context in particular, there has been limited empirical understanding of the specific effect of agility elements on software development effectiveness. By drawing on existing literature on ISD agility, which suggests that agility is composed of multiple facets, and adopting the AHP technique, this study contributes by empirically examining practitioners' views surrounding the relative importance of these different facets of agility with respect to two different measures of success of a DSD project. Our study delivers three major findings.

First, through the examination of the relative importance of the agility facets, the study initiates the first step towards the eventual development of 'EP theories' (theories for explaining and predicting) surrounding agility in distributed ISD, articulating 'theoretical constructs and

[suggesting ...] relationships among them' (Gregor, 2006, p. 626). Over the last few years, there have been calls within the IS discipline for researchers to focus not only on new concepts, but also on how the concepts work together toward a particular dependent variable.

Second, the study illustrates the use of the AHP technique, which has a rich tradition in decision sciences, but remains underutilized within the information systems discipline. Given that most organizational decision-making (e.g., adoption of a new Enterprise Resource Planning (ERP) system) involves trade-offs between multiple factors, use of this technique can create a useful assessment framework, and thereby guide decision-makers. Our study encourages future researchers within IS to apply this technique in examining other IS-related decision-making contexts.

Third, by empirically unearthing the relative importance of the different facets of agility in enhancing ISD success, the study's results provide guidance to organizational decision-makers regarding the specific types of agility they should focus on in an effort to enhance the chances of their DSD project's success. In other words, the facets framework is useful for assigning greater priority/attention to those facets found to be substantially contributing toward a goal. Given that most distributed ISD projects are constrained by budget, time, and other resources issues, by using this framework, decision-makers can direct more focused attention to the facets of agility that are most relevant, as opposed to directing resources toward an arbitrary set of agility facets, or wastefully toward all facets.

To conclude, incorporating agility considerations in ISD appears to be a growing trend within the research and practitioner communities, and we hope that, through our investigation, we are able to provide knowledge that is potentially helpful in managing DSD teams.

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